ASSIGNMENT 2

Subject: Data Structures and Algorithms **Subject Code: BCSC 0006**

Last Date of Submission: April 20, 2023 **Max Marks: 50**

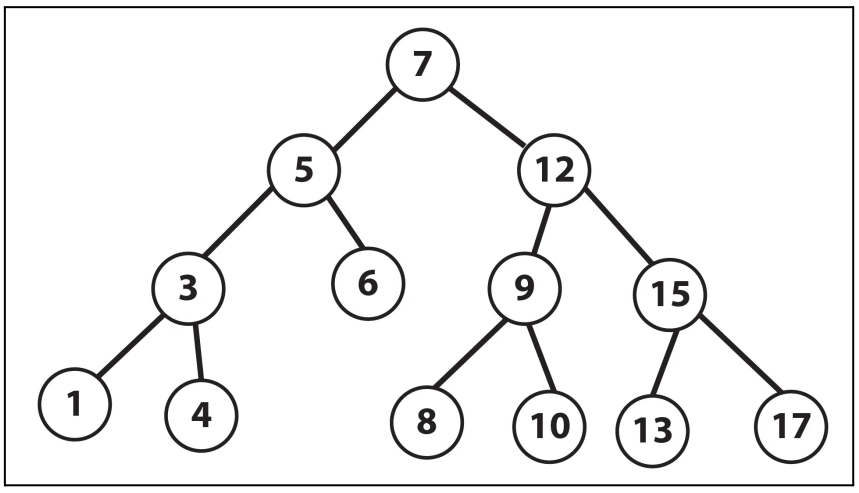
Instructions:

- Assignment should be handwritten.
- Will Only accept on portal
- Handwriting should be neat and clean
- Assignment will only be accepted before the deadline, not after it in any circumstances.
- All questions are compulsory.
- Mention Your Name and University Roll No. on the top of each page of solution.

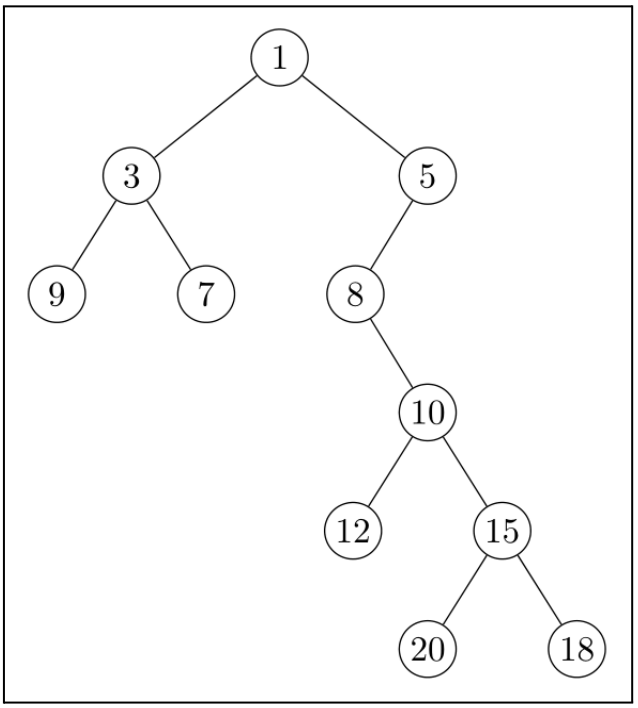
Numericals Type (3 x 10)

1. Create a BST for the following set of values show all the steps involved:
   a. 8, 3, 1, 10, 14, 6, 4, 13, 7
   b. K, A, F, S, L, Z, I, J, B, D
2. Create a AVL Tree for the following set of values show all the steps involved:
   a. 50 , 20 , 60 , 10 , 8 , 15 , 32 , 46 , 11 , 48
   b. H, I, J, B, A, E, C, F, D, G, K, L

3. Delete the key from the following Tree, show all the steps involved:
   a. Key = 9

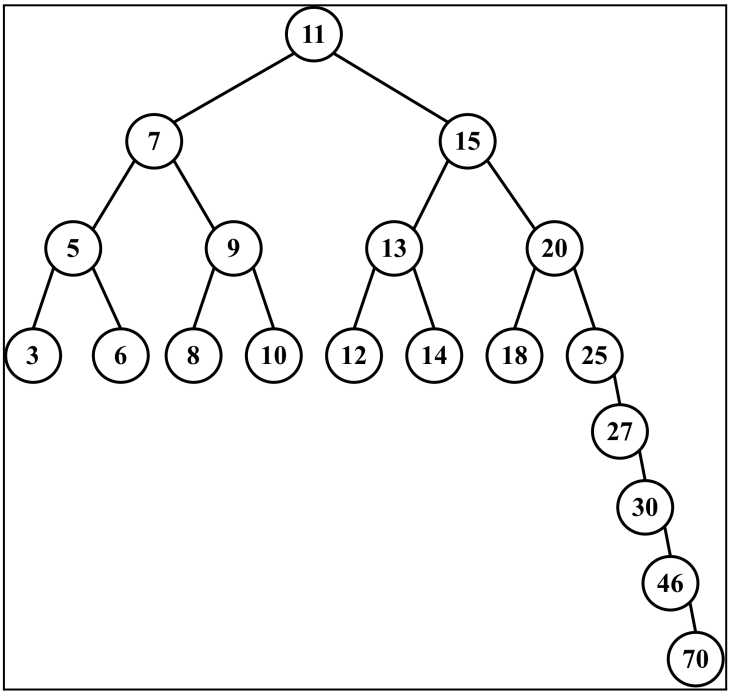


b. Key = 10

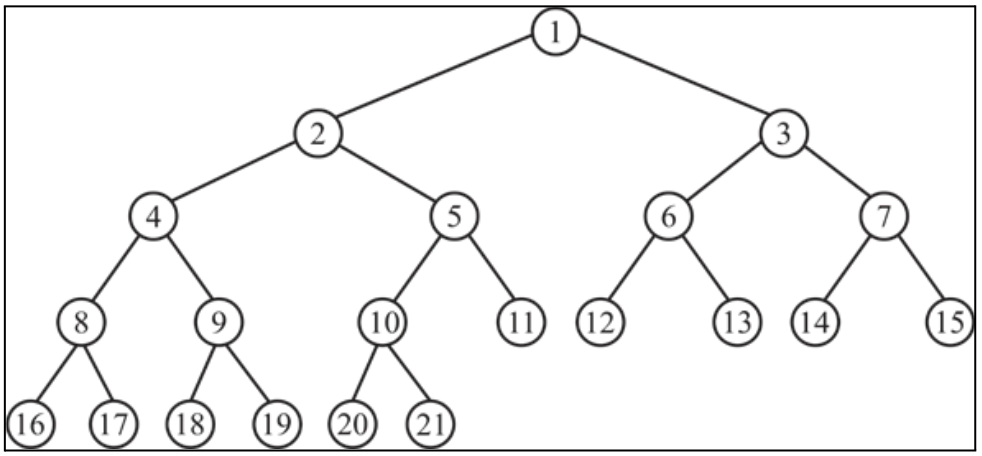


4. Create the **LevelOrder, InOrder, PreOrder, PostOrder** Traversal of the following the Tree:

a.



b.



5. Sort the given data in array using **Merge Sort , Quick Sort and Heap Sort**, all the steps involved of Sorting. Compare the **Time Complexity Anaylsis** of each type of Sort for data in this array.

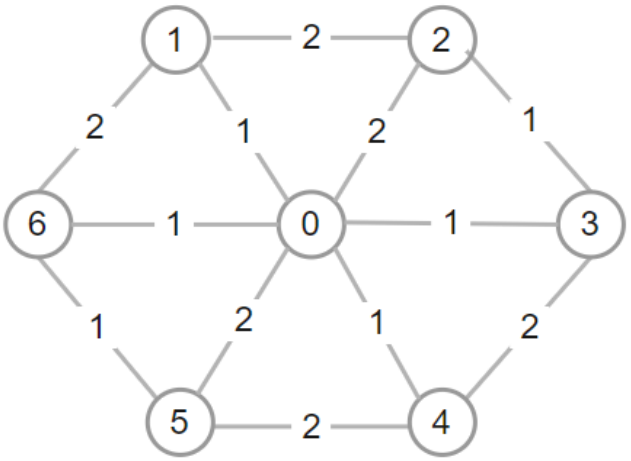
81	89	9	11	14	76	54	22

6. Sort the given data in array using **insertion sort**, and explain all the steps involved in it:

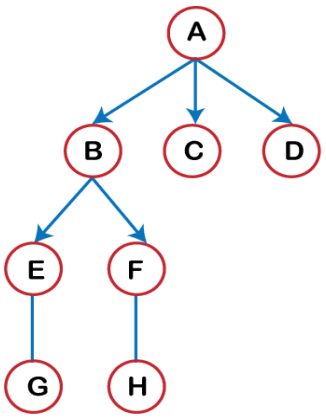
-5, -8, -2, 0, -4, -1, -6

7. Delete 48 and 11 from the AVL tree, as per the given data in Q2(a).

8. Find the minimum spanning tree using Kruskal's algorithm for the following graph:



9. Find BFS and DFS for the following graph and show all the steps involved:



10. Solve 4x4 queen problem with the help of Backtracking.

Code Type (4 x 5)

1. Write the Program for Iterative searching in Binary Search Tree.
2. Write the Program to implement the Search a element in Binary Tree with the condition that if element/key is not found in the tree, insert the key/element in the tree.
3. Write a java program to sort the elements by using selection sort in array from the given binary tree.
4. Write a recursive program to solve 3-peg based tower of hanoi problem.
5. Write a program to reverse printing of each node's data element in Singly Linked List. For eg: Linked list having elements 1-2-3-4 where 1 is head node then print 4-3-2-1